A REMARK ON GRADIENTS OF HARMONIC FUNCTIONS IN DIMENSION ≥ 3
A BOURGAIN-PEDIA DIGESTION

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§0. Editorial preface

This document is the Bourgain-pedia *digestion* of the Bourgain–Wolff note *A remark on gradients of harmonic functions in dimension* ≥ 3. It is that paper at full length: every step the authors compressed is written out, every argument they omitted is supplied, and nothing else has been changed. It is not a survey, not a modernisation, and not a better proof. Where a modern route exists it is confined to §9.

Read it beside the transcription `...-original.tex`, which reproduces the preprint as printed, including its slips. Section and lemma numbering here is the authors'.

How to read the boxes. Four kinds of material are ours, and each is boxed so that you can see at a glance where the authors stop and we start: *expansion* (a step of theirs, written out), *filled* (an argument absent from the original), *our notation*, and *caution* (a step we could not close). Unboxed prose and displays are theirs.

Source. The IHES preprint (September 1989), obtained from the IHES preprint fonds. The journal version appeared in 1990 and may differ after refereeing; every claim here is a claim about the preprint. [F4]

Five discrepancies in the original, each verified against the scan at 320 dpi and each repaired below with the repair marked:

(i) p. 4: $\int_{|x|>1} |d\rho_\varepsilon/dn| \leq C\varepsilon\delta^{-p}$ is printed; the estimate $(*)$ yields exponent -1, not $-p$. Both give the conclusion. [B23]

(ii) p. 5: the first display of Lemma 2's proof prints $|1 + d\widehat{F}_\varepsilon/dn|^p - |1 + d\widehat{F}_\varepsilon/dn|^p$, identically 0; the first term must be $|I + d\widehat{F}_\varepsilon/dn|^p$. [C4]

(iii) p. 9: the final parenthesis of Lemma 4 prints $e^{+\frac{1}{2}p\beta}$, which is > 1 and contradicts the sentence after it; the recomputation forces $e^{-\frac{1}{2}p\beta}$. [D21]

(iv) p. 10: condition (2) prints ε_{n+1}^{d-1}; the use made of it requires $\varepsilon_{n+1}^{(d-1)/2}$. [D23]

(v) p. 10: $|Q| < K_{n+1}^{-p} e^{-\beta pn} \int_Q |du_n/dn|^p$ is printed; the algebra gives $e^{+\beta pn}$, and the next display on p. 11 uses $+$. [D31]

None of the five affects the theorem. Two further points needed real repair rather than a sign change, and are treated where they arise: the lower bound for $|1 + d\widehat{G}_\varepsilon/dn|$ fails exactly at $|x| = 1$ (§2.6, §3.3), and the final passage from "$f = 0$ and $\partial_n f = 0$" to "$\nabla f = 0$" is not in the paper at all (§7.5).

What we could not close. One item, and it is the load-bearing one: the Alexandrov–Kargaev inequality imported as reference [1] (§2.2). Its source is "private communication (to appear)" and we have not obtained it. Everything else in the paper is proved here.

Date: Digestion generated August 24, 2026.

§0′. Notation

The authors' notation.

- $\mathbb{R}^d_+ = \{x = (x', x_d) : x' \in \mathbb{R}^{d-1},\ x_d > 0\}$, boundary identified with $\mathbb{R}^{d-1}$.
- $D(0, r) \subset \mathbb{R}^{d-1}$ is the open ball of radius r; $|E|$ is Lebesgue measure on $\mathbb{R}^{d-1}$.
- $Q(N) = [-\frac{N}{2}, \frac{N}{2}]^{d-1}$; in particular $Q(1)$ is the unit cube. $\ell(Q)$ is the side length of a cube Q and a_Q its centre.
- $\widehat{g}$ is the harmonic extension of $g : \mathbb{R}^{d-1} \to \mathbb{R}$ to $\mathbb{R}^d_+$. From §4 on the authors drop the hat and let g denote both. [F2]
- d/dn is the normal derivative on $\{x_d = 0\}$ of a harmonic extension; ∇_T is differentiation in the $\mathbb{R}^{d-1}$ directions.
- $\mathcal{H}_n$: the $\delta_n^{-(d-1)}$ closed cubes of side δ_n tiling $Q(1)$. $\mathcal{Q}_n \subset \mathcal{H}_n$, and $V_n = \bigcup\{Q : Q \in \mathcal{Q}_n\}$.
- e_d is the unit vector in the x_d direction.

OUR NOTATION — not in the original

- ∂_n **for** d/dn, and we fix the convention the paper never states:
$$\partial_n g(x') := \frac{\partial \widehat{g}}{\partial x_d}(x', 0^+),$$
the derivative along the *inward* normal $+e_d$. §2.1 shows this is the convention the paper's estimates are written in. [A4]
- $A \lesssim B$ means $A \le CB$ with C depending only on d, on p, and on the fixed parameter N once it is chosen — never on $\varepsilon, \delta, \lambda, j, n$, nor on $K_n, \varepsilon_n, \delta_n$. Any other dependence is written out. The paper uses $\lesssim$ without ever saying this. [F1]
- $\|g\|_p := (\int |g|^p)^{1/p}$ **for** $0 < p < 1$**.** This is *not* a norm and obeys no triangle inequality; every step that looks like one is justified by Background instead. [F3]
- P for the Poisson kernel of $\mathbb{R}^d_+$ (Background).
- $s = |x'|/\varepsilon$, the natural variable in §2.

§0′′. Background

Everything the paper imports, stated in full. The authors state none of these.

Background (Poisson kernel and harmonic extension)**.** For $x' \in \mathbb{R}^{d-1}$, $t > 0$ put $P(x', t) = c_d\, t\, (|x'|^2 + t^2)^{-d/2}$ with $c_d = \Gamma(d/2)\pi^{-d/2}$, so that $\int_{\mathbb{R}^{d-1}} P(x', t)\, dx' = 1$ for every $t > 0$. For bounded continuous g the harmonic extension is $\widehat{g}(x', t) = (P(\cdot, t) * g)(x')$; it is harmonic in $\mathbb{R}^d_+$ and $\widehat{g}(\cdot, t) \to g$ as $t \downarrow 0$.

Background (Dirichlet-to-Neumann operator)**.** With $\widehat{g}(\xi, t) = e^{-|\xi| t}\hat{g}(\xi)$ on the Fourier side,
$$\partial_n g = \frac{\partial \widehat{g}}{\partial x_d}\Big|_{x_d = 0^+} = -|\nabla'| g, \qquad \widehat{|\nabla'| g}(\xi) = |\xi| \hat{g}(\xi).$$
Since $|\nabla'| = \sum_{k=1}^{d-1} R_k \partial_k$ with R_k the Riesz transforms on $\mathbb{R}^{d-1}$ (multipliers $-i\xi_k/|\xi|$),
$$\partial_n g = -\sum_{k=1}^{d-1} R_k\, \partial_k g. \tag{1}$$

FILLED — an argument omitted in the original, supplied here

(1) is *the* reason the Riesz transforms appear in Lemma 1, and the paper never says so: it writes "By $L^1 \to$ weak-L^1 and Hölder estimates for the Riesz transforms we have…" with no

indication of what the Riesz transforms are being applied to. They are applied to $\nabla_T \rho_\varepsilon^j$, and (1) is what converts a bound on the *tangential* gradient of a boundary function into a bound on the *normal* derivative of its harmonic extension. [B13]

Background (Riesz transforms: the two estimates used)**.** On $\mathbb{R}^{d-1}$: (i) each R_k is of weak type $(1,1)$: $|\{|R_k h| > \lambda\}| \le C_d \lambda^{-1} \|h\|_1$; (ii) for $0 < \alpha < 1$ each R_k is bounded on the homogeneous Hölder space C^α: $\|R_k h\|_{C^\alpha} \le C_{d,\alpha} \|h\|_{C^\alpha}$.

Background (Representation away from the support)**.** If $h \in C_c^\infty(\mathbb{R}^{d-1})$ and $x \notin \operatorname{supp} h$ then $\partial_n h(x) = c_d' \int_{\mathbb{R}^{d-1}} |x-y|^{-d} h(y)\, dy$ with $c_d' = -c_d\,(d-1) \int \cdots$; only $|c_d'| \lesssim 1$ is used. This is (1) written as a singular integral, legitimate off the support because the kernel is then smooth.

Background (p-subadditivity, $0 < p \le 1$)**.** $(a+b)^p \le a^p + b^p$ for $a, b \ge 0$; consequently, for all real a, b,

$$\big| |1+a|^p - |1+b|^p \big| \le |a-b|^p. \tag{2}$$

FILLED — an argument omitted in the original, supplied here

Proof of Background . For $0 < p \le 1$ the function $t \mapsto t^p$ is concave on $[0, \infty)$ with $0^p = 0$, hence subadditive: for $a, b \ge 0$, $\frac{a}{a+b}(a+b)^p + \frac{b}{a+b}(a+b)^p = (a+b)^p$ while concavity through the origin gives $a^p \ge \frac{a}{a+b}(a+b)^p$ and $b^p \ge \frac{b}{a+b}(a+b)^p$; adding yields $a^p + b^p \ge (a+b)^p$. For (2), put $u = |1+a|$, $v = |1+b|$ and assume $u \ge v$. Then $u \le v + |a-b|$ by the triangle inequality, so $u^p \le (v + |a-b|)^p \le v^p + |a-b|^p$, i.e. $u^p - v^p \le |a-b|^p$. ☐ [B19]

Background (Mean value step)**.** If $|1+a| \ge c > 0$ and $|1+b| \ge c > 0$ then $\big||1+a|^p - |1+b|^p\big| \le p\, c^{p-1} |a-b|$.

FILLED — an argument omitted in the original, supplied here

Proof. $t \mapsto |t|^p$ is C^1 away from 0 with derivative of modulus $p|t|^{p-1}$, which for $0 < p < 1$ is decreasing in $|t|$; on the segment joining $1+a$ to $1+b$ — which stays in $\{|t| \ge c\}$ when $1+a$ and $1+b$ have the same sign — the derivative is at most pc^{p-1}. If $1+a$ and $1+b$ have opposite signs the segment crosses 0; but then $|a-b| \ge 2c$ and the claim follows from $\big||1+a|^p - |1+b|^p\big| \le \max(|1+a|, |1+b|)^p$ together with $|1+a|, |1+b| \le |a-b| + c \le \frac{3}{2}|a-b|$. ☐ [B22]

Background (Density points)**.** If $E \subset \mathbb{R}^{d-1}$ is measurable, a.e. $x \in E$ is a point of density 1 of E. If moreover $g \in C^1$ and $g \equiv 0$ on E, then $\nabla g(x) = 0$ at every density point x of E.

FILLED — an argument omitted in the original, supplied here

Proof of the second sentence. Let x be a density point of E and suppose $\nabla g(x) = v \ne 0$. Pick the unit vector $e = v/|v|$. By differentiability, $g(x + ry) = r\langle v, y\rangle + o(r)$ uniformly for $|y| \le 1$. The set $S = \{y : |y| \le 1,\ \langle v, y\rangle > \frac{1}{2}|v|\}$ is a spherical cap of positive measure κ, and for r small $g > 0$ on $x + rS$, so $(x + rS) \cap E = \emptyset$. Then $|B(x,r) \setminus E| \ge \kappa r^{d-1}$, contradicting density 1. ☐ [D34]

Background (Maximum principle for the gradient)**.** If v is harmonic and bounded on $\mathbb{R}^d_+$, continuous on $\overline{\mathbb{R}^d_+}$, and $v(x) \to 0$ as $|x| \to \infty$, then $\sup_{\mathbb{R}^d_+} |v| = \sup_{\mathbb{R}^{d-1}} |v|$. Each partial derivative of a harmonic function is harmonic, so the same holds for each $\partial_j v$ provided it too is bounded and decays.

§1. The theorem

Theorem (Bourgain–Wolff). *If $d \geq 3$ there is a harmonic function $f : \mathbb{R}^d_+ \to \mathbb{R}$ which is C^1 up to the boundary and such that f and ∇f vanish on a common boundary set with positive measure.*

OUR NOTATION — not in the original

The original prints "C'" for C^1. [A1]

FILLED — an argument omitted in the original, supplied here

What the statement must mean, and one thing it must exclude. [A2, A3]
f is harmonic in the open half space and extends to a C^1 function on $\overline{\mathbb{R}^d_+}$; "$f$ vanishes" refers to its boundary values, and "∇f vanishes" to the full gradient of that C^1 extension, evaluated on the boundary. So the assertion is: there is $E \subset \mathbb{R}^{d-1}$ with $|E| > 0$ and $f|_E = 0$, $\nabla f|_E = 0$. As literally stated the theorem is satisfied by $f \equiv 0$. The construction does better, and nontriviality has to be read into it: we record here that the construction produces $f \not\equiv 0$, and where. In §4 the initial datum u_0 is a smooth function on $\mathbb{R}^{d-1}$ vanishing on $Q(1)$; if $u_0 \not\equiv 0$ then, since every later modification is supported inside $Q(1)$ (§4.4), $f = u_0$ on $\mathbb{R}^{d-1} \setminus Q(1)$ and hence $f \not\equiv 0$. *So: choose u_0 smooth, vanishing on $Q(1)$, and not identically zero.* The paper does not say this.

Outline of proof (theirs). Start with a function u_0 which vanishes on an open subset of the boundary. Using a successive modification ("correction theorem") procedure, decrease the normal derivative to 0 on a subset with large measure. To get the theorem it is necessary to keep the original function unchanged on a set with large measure. Thus the functions added on during the modification procedure must have small compact support. The idea for constructing those functions is due to A.B. Alexandrov and P. Kargaev [1].

EXPANSION — a step he compressed, written out

The shape of the argument, in one paragraph. The whole proof is a contest between two budgets. Modifying u_n on a cube Q multiplies the local L^p size of $\partial_n u_n$ by a definite factor $e^{-2\beta} < 1$ (Lemma 2) — that is the gain. The modification is not free: it changes u_n itself on a disc inside Q, and it perturbs $\partial_n u_n$ on *other* cubes (Lemma 3 controls the interference). The gain is geometric in n; the two costs are $\sum \varepsilon_{n+1}^{(d-1)/2}$ (measure where f moved) and $\sum K_{n+1}^{-p}$ (measure discarded from V_n). Condition (2) of §7 says the two costs together come to less than $|Q(1)| = 1$, so something is left over, and that leftover is the set E.

§2. Lemma 1: the Alexandrov–Kargaev building block

Lemma 1. *If $p > 0$ is small enough then for all sufficiently small ε there is $F_\varepsilon : \mathbb{R}^{d-1} \to \mathbb{R}$ with* $\operatorname{supp} F_\varepsilon \subset D(0, \varepsilon^{1/2})$ *and, $\widehat{F}_\varepsilon$ denoting its harmonic extension to $\mathbb{R}^d_+$,*

$$\int_{\mathbb{R}^{d-1}} \Big(\big|1 + \partial_n F_\varepsilon\big|^p - 1 \Big)\, dx < -\eta \tag{3}$$

with $\eta > 0$ independent of ε; moreover

$$|\nabla \widehat{F}_\varepsilon| \lesssim \min(\varepsilon^{-d}, |x|^{-d}) \quad \text{on } \mathbb{R}^{d-1}. \tag{4}$$

OUR NOTATION — not in the original

The original prints $|x| \subset \varepsilon^{1/2}$ for $|x| < \varepsilon^{1/2}$ [B1], and in (4) prints a lower-case $\widehat{f}_\varepsilon$ where $\widehat{F}_\varepsilon$ is meant, as its own restatement on p. 4 confirms [B2].

FILLED — an argument omitted in the original, supplied here

Quantifier order. [B3] The statement is: there is $p_0 = p_0(d) > 0$ such that for every $p \in (0, p_0)$ there are $\eta = \eta(p, d) > 0$ and $\varepsilon_0 = \varepsilon_0(p, d) > 0$ with the following property: for every $\varepsilon \in (0, \varepsilon_0)$ there exists F_ε as described. The point of "η independent of ε" is that η may depend on p and d but not on ε; §2.7 shows where that matters. §2.5 will force $p < (d-1)/d$, so $p_0 \leq (d-1)/d$.

2.1 The Alexandrov–Kargaev function, and the sign convention. Alexandrov–Kargaev use the function $G_\varepsilon : \mathbb{R}^d_+ \to \mathbb{R}$,

$$G_\varepsilon(x) = \frac{-(\varepsilon + x_d)}{|x + \varepsilon e_d|^d}. \tag{5}$$

EXPANSION — a step he compressed, written out

G_ε is harmonic on a neighbourhood of $\overline{\mathbb{R}^d_+}$. [B5] By Background , $P(x', t) = c_d t(|x'|^2 + t^2)^{-d/2}$, so

$$G_\varepsilon(x) = -\frac{1}{c_d} P(x', x_d + \varepsilon),$$

i.e. G_ε is $-1/c_d$ times the Poisson kernel translated *down* by ε. The Poisson kernel is harmonic on the open upper half space, so G_ε is harmonic on $\{x_d > -\varepsilon\}$, which contains $\overline{\mathbb{R}^d_+}$. In particular G_ε is real-analytic up to and across the boundary $\{x_d = 0\}$, and $\partial_n G_\varepsilon$ below is an honest classical derivative, not a boundary limit.

EXPANSION — a step he compressed, written out

The normal derivative, computed exactly. Differentiating (5) in x_d and setting $x_d = 0$, with $r = (|x'|^2 + \varepsilon^2)^{1/2}$:

$$\partial_n G_\varepsilon(x') = \frac{\partial G_\varepsilon}{\partial x_d}(x', 0) = -r^{-d} + d\,\varepsilon^2 r^{-d-2} = -r^{-d-2}\big(r^2 - d\varepsilon^2\big).$$

In the scaled variable $s = |x'|/\varepsilon$, so that $r = \varepsilon(s^2+1)^{1/2}$,

$$\partial_n G_\varepsilon = -\,\varepsilon^{-d}\, h(s), \qquad h(s) = (s^2+1)^{-\frac{d+2}{2}}\big(s^2 + 1 - d\big). \tag{6}$$

Three features of (6) drive everything below:

$$\partial_n G_\varepsilon(0) = (d-1)\varepsilon^{-d} > 0, \qquad \partial_n G_\varepsilon = 0 \text{ at } |x'| = \varepsilon\sqrt{d-1}, \qquad \partial_n G_\varepsilon \sim -|x'|^{-d} \;\; (|x'| \gg \varepsilon).$$

FILLED — an argument omitted in the original, supplied here

Why the inward normal. [A4] The paper never fixes the sign of d/dn, and the two choices are not interchangeable here. With the inward convention $\partial_n = \partial_{x_d}|_{x_d=0}$ used above, $\partial_n G_\varepsilon \sim -|x'|^{-d}$ in the far field, so $1 + \partial_n G_\varepsilon \sim 1 - |x'|^{-d}$ passes through 0 near $|x'| = 1$ and is < 1 beyond it — and it is precisely the region $|x'| \gtrsim 1$, where $|1 + \partial_n G_\varepsilon|^p - 1 \approx -p|x'|^{-d} < 0$, that can make the integral (7) negative. With the outward convention every regime contributes a positive amount and (7) is false. So the inward normal is the paper's convention, and we adopt it.

2.2 The imported inequality. As Alexandrov and Kargaev show,

$$\int_{\mathbb{R}^{d-1}} \Big(\big|1 + \partial_n G_\varepsilon\big|^p - 1 \Big) dx < -2M < 0 \tag{7}$$

for p below a certain critical number and ε sufficiently small.

CAUTION — not fully justified here; OPEN-GAP in the ledger

This is the one thing in the paper we cannot verify. [B4] Reference [1] is "A.B. Aleksandrov, P. Kargaev, *Private communication* (to appear)", and we have not obtained it; the published form is not identified [E4]. Everything below is therefore *conditional on* (7), which we treat as a hypothesis. Status: **OPEN-GAP**.
What can be said from here. Substituting $x = \varepsilon y$ in (7) and using (6), the integral equals $\varepsilon^{d-1} \int_{\mathbb{R}^{d-1}} (|1 - \varepsilon^{-d} h(|y|)|^p - 1)\, dy$, and splits into three regimes:

- $|x'| \lesssim \varepsilon$: $|1 + \partial_n G_\varepsilon| \approx (d-1)\varepsilon^{-d}$, contributing $O(\varepsilon^{d-1-dp})$ — positive, $\to 0$.
- $\varepsilon \ll |x'| < 1$: $|1 + \partial_n G_\varepsilon| \approx |x'|^{-d} \gg 1$, giving $\int_0^1 (r^{-dp} - 1) r^{d-2} dr = \frac{dp}{(d-1)(d-1-dp)}$ — positive, and $O(p)$.
- $|x'| > 1$: $1 + \partial_n G_\varepsilon \approx 1 - |x'|^{-d} \in (0, 1)$, giving $\approx -p \int_{|x'|>1} |x'|^{-d} dx'$ — *negative*, and $O(p)$; together with the annulus around the zero at $|x'| \approx 1$, where the integrand is close to -1.

So (7) asserts that at small p the third regime beats the second. Both are $O(p)$, so the inequality is a genuine computation about the constants and not something the shape of the estimate settles. We do not reproduce it.

OUR NOTATION — not in the original

The proof of Lemma 1 uses η where (7) produced M; the two are identified, $\eta := M$. [B6]

2.3 The cut-off, and why $\operatorname{supp} F_\varepsilon$ **is small.** Let $\delta = \varepsilon^{1/2}$ and for $j = 0, 1, 2, \ldots$ let $\psi_j : \mathbb{R}^{d-1} \to \mathbb{R}$ be functions with

$$\operatorname{supp} \psi_j \subset \{2^{j-1}\delta \le |x| \le 2^{j+1}\delta\}, \qquad \sum_j \psi_j = 1 \ \text{ for } |x| > \delta, \qquad |\nabla^k \psi_j| \lesssim (2^j \delta)^{-k}.$$

Put $\psi = \sum_j \psi_j$, $\rho_\varepsilon^j = \psi_j G_\varepsilon$, $\rho_\varepsilon = \sum_j \rho_\varepsilon^j = \psi G_\varepsilon$, and

$$F_\varepsilon = G_\varepsilon - \rho_\varepsilon = (1 - \psi) G_\varepsilon.$$

FILLED — an argument omitted in the original, supplied here

Such a family exists. [B7] Fix $\chi \in C_c^\infty(\mathbb{R})$ with $\operatorname{supp} \chi \subset [\frac{1}{2}, 2]$, $\chi \ge 0$, and $\sum_{j \in \mathbb{Z}} \chi(2^{-j} t) = 1$ for $t > 0$ (a standard dyadic partition: take any $\theta \in C_c^\infty$ with $\theta = 1$ on $[0, 1]$, $\theta = 0$ on $[2, \infty)$, and set $\chi(t) = \theta(t) - \theta(2t)$). Put $\psi_j(x) = \chi\big(|x|/(2^j\delta)\big)$ for $j \ge 0$. Then $\operatorname{supp} \psi_j \subset \{2^{j-1}\delta \le |x| \le 2^{j+1}\delta\}$; $\sum_{j \ge 0} \psi_j(x) = 1$ once $|x| > \delta$, because the omitted terms $j < 0$ are supported in $\{|x| < \delta\}$; and $|\nabla^k \psi_j| \le \|\chi^{(k)}\|_\infty (2^j\delta)^{-k}$ by the chain rule.

EXPANSION — a step he compressed, written out

$\operatorname{supp} F_\varepsilon \subset D(0, \varepsilon^{1/2})$**, and** F_ε **is smooth.** [B8] With the ψ_j above, $\psi = 0$ on $\{|x| < \delta/2\}$ (every ψ_j, $j \ge 0$, vanishes there) and $\psi = 1$ on $\{|x| > \delta\}$. Hence $1 - \psi$ is supported in $\overline{D(0, \delta)}$, and since $F_\varepsilon = (1 - \psi) G_\varepsilon$,

$$\operatorname{supp} F_\varepsilon \subset \overline{D(0, \delta)} = \overline{D(0, \varepsilon^{1/2})},$$

which is the claim in Lemma 1. Smoothness: on the boundary $G_\varepsilon(x',0) = -\varepsilon(|x'|^2+\varepsilon^2)^{-d/2}$ is real-analytic (the denominator never vanishes, as $\varepsilon > 0$), and $1-\psi$ is smooth; so $F_\varepsilon \in C_c^\infty(\mathbb{R}^{d-1})$. Note in passing that F_ε agrees with G_ε on $\{|x| < \delta/2\}$: the cut-off removes the far field, not the singular core.

2.4 Estimates on ρ_ε. Let ∇_T be differentiation in the $\mathbb{R}^{d-1}$ directions. Then $|\nabla_T^k G_\varepsilon| \lesssim \varepsilon|x|^{-(d+k)}$ for $k = 0,1,2$; therefore $|\nabla_T^k \rho_\varepsilon^j| \lesssim \varepsilon(2^j\delta)^{-(d+k)}$, whence

$$\|\nabla_T \rho_\varepsilon^j\|_1 \lesssim \varepsilon(2^j\delta)^{-2}, \qquad \|\nabla_T \rho_\varepsilon^j\|_{C^\alpha} \lesssim \varepsilon(2^j\delta)^{-(d+1+\alpha)}.$$

EXPANSION — a step he compressed, written out

The three estimates, done. [B9, B10, B11, B12] On the boundary, $G_\varepsilon(x',0) = -\varepsilon(|x'|^2+\varepsilon^2)^{-d/2}$.

- $k = 0$: $|G_\varepsilon| = \varepsilon(|x'|^2+\varepsilon^2)^{-d/2} \le \varepsilon|x'|^{-d}$, with constant 1.
- $k = 1$: $\nabla_T G_\varepsilon = d\,\varepsilon\, x'(|x'|^2+\varepsilon^2)^{-\frac{d}{2}-1}$, so $|\nabla_T G_\varepsilon| \le d\,\varepsilon|x'|\cdot|x'|^{-d-2} = d\,\varepsilon|x'|^{-(d+1)}$.
- $k = 2$: differentiating once more produces two terms, of sizes $d\varepsilon(|x'|^2+\varepsilon^2)^{-\frac{d}{2}-1}$ and $d(d+2)\varepsilon|x'|^2(|x'|^2+\varepsilon^2)^{-\frac{d}{2}-2}$, both $\le C_d\,\varepsilon|x'|^{-(d+2)}$.

Note the paper writes "$\le$" where the constant depends on d; we write $\lesssim$.
For $\rho_\varepsilon^j = \psi_j G_\varepsilon$, Leibniz gives $\nabla_T^k\rho_\varepsilon^j = \sum_{i\le k}\binom{k}{i}(\nabla_T^i\psi_j)(\nabla_T^{k-i}G_\varepsilon)$, and on $\operatorname{supp}\psi_j$ we have $|x| \sim 2^j\delta$, so

$$|\nabla_T^k\rho_\varepsilon^j| \lesssim \sum_{i\le k}(2^j\delta)^{-i}\cdot\varepsilon(2^j\delta)^{-(d+k-i)} \lesssim \varepsilon(2^j\delta)^{-(d+k)}.$$

For the L^1 bound, $\operatorname{supp}\nabla_T\rho_\varepsilon^j$ lies in an annulus of measure $\lesssim (2^j\delta)^{d-1}$, so

$$\|\nabla_T\rho_\varepsilon^j\|_1 \lesssim \varepsilon(2^j\delta)^{-(d+1)}\cdot(2^j\delta)^{d-1} = \varepsilon(2^j\delta)^{-2}.$$

For the Hölder bound, fix once and for all an $\alpha \in (0,1)$ (any choice works; $\alpha = \frac{1}{2}$ will do). A function with $|\nabla_T\rho^j| \lesssim \varepsilon(2^j\delta)^{-(d+1)}$ and $|\nabla_T^2\rho^j| \lesssim \varepsilon(2^j\delta)^{-(d+2)}$ satisfies $\|\nabla_T\rho^j\|_{C^\alpha} \lesssim \|\nabla_T\rho^j\|_\infty^{1-\alpha}\|\nabla_T^2\rho^j\|_\infty^\alpha \lesssim \varepsilon(2^j\delta)^{-(d+1+\alpha)}$, which is the stated bound.

By (1) and Background applied to $h = \partial_k\rho_\varepsilon^j$,

$$\left|\left\{x : \left|\partial_n\rho_\varepsilon^j(x)\right| > \lambda\right\}\right| \lesssim \varepsilon(2^j\delta)^{-2}\lambda^{-1}, \tag{8}$$

$$\left\|\partial_n\rho_\varepsilon^j\right\|_{C^\alpha} \lesssim \varepsilon(2^j\delta)^{-(d+1+\alpha)}. \tag{9}$$

EXPANSION — a step he compressed, written out

Near field: $|\partial_n\rho_\varepsilon^j(x)| \lesssim \varepsilon(2^j\delta)^{-(d+1)}$ for $|x| < 2\cdot 2^{j+1}\delta$. [B15] Write $R = 2\cdot 2^{j+1}\delta$, $T = \varepsilon(2^j\delta)^{-(d+1)}$, and $g = \partial_n\rho_\varepsilon^j$.
By (8) with $\lambda = T$, the set $\{|g| > T\}$ has measure $\lesssim \varepsilon(2^j\delta)^{-2}T^{-1} = (2^j\delta)^{d-1}$. The ball $D(0,R)$ has measure $c_dR^{d-1} = c_d\,4^{d-1}(2^j\delta)^{d-1}$. So if the implicit constant in (8) is C_1, then applying (8) at level $\lambda = KT$ instead, with K a constant chosen so that $C_1K^{-1} < c_d4^{d-1}$, gives

$$|\{|g| > KT\}\cap D(0,R)| < |D(0,R)|,$$

so there exists $x_0 \in D(0,R)$ with $|g(x_0)| \le KT$. That is the "*some* x" of the original.
To pass to *all* $x \in D(0,R)$, use (9): for any $x \in D(0,R)$,

$$|g(x)| \le |g(x_0)| + \|g\|_{C^\alpha}|x-x_0|^\alpha \le KT + C_2\,\varepsilon(2^j\delta)^{-(d+1+\alpha)}(2R)^\alpha.$$

Since $2R = 8 \cdot 2^j\delta$, the second term is $C_2 8^\alpha \varepsilon(2^j\delta)^{-(d+1)} = C_2 8^\alpha T$. Hence $|g(x)| \le (K + C_2 8^\alpha)T \lesssim T$ throughout $D(0,R)$, as claimed. The constants depend only on d and α, both fixed.

Furthermore, if $|x| > 2 \cdot 2^{j+1}\delta$ then, by Background ,

$$\left|\partial_n \rho_\varepsilon^j(x)\right| = \left|c_d' \int_{|y|<2^{j+1}\delta} |x-y|^{-d} \rho_\varepsilon^j(y)\,dy\right| \lesssim \varepsilon(2^j\delta)^{-1}|x|^{-d}, \tag{10}$$

using $|\rho_\varepsilon^j| \lesssim \varepsilon(2^j\delta)^{-d}$.

EXPANSION — a step he compressed, written out

Where (10) comes from. [B16] Since $|x| > 2\cdot 2^{j+1}\delta$ and $|y| < 2^{j+1}\delta$ we have $|x-y| \ge \frac{1}{2}|x|$, so $|x-y|^{-d} \le 2^d|x|^{-d}$ and the integral is at most $2^d|x|^{-d}\|\rho_\varepsilon^j\|_1$. Now $\|\rho_\varepsilon^j\|_1 \lesssim \varepsilon(2^j\delta)^{-d}\cdot(2^j\delta)^{d-1} = \varepsilon(2^j\delta)^{-1}$, the annulus again having measure $\lesssim (2^j\delta)^{d-1}$. This gives (10). The representation itself is legitimate because x lies outside $\operatorname{supp}\rho_\varepsilon^j$, where the kernel of $|\nabla'|$ is smooth.

In other words

$$\left|\partial_n \rho_\varepsilon^j\right| \lesssim \varepsilon(2^j\delta)^{-1}\min\!\big(|x|^{-d}, (2^j\delta)^{-d}\big) \quad \text{on } \mathbb{R}^{d-1}, \tag{11}$$

and summing over j,

$$\left|\partial_n \rho_\varepsilon\right| \lesssim \varepsilon\,\delta^{-1}\min\!\big(\delta^{-d}, |x|^{-d}\big). \tag{$*$}$$

EXPANSION — a step he compressed, written out

The summation. [B17] Fix x and let j_0 be the least j with $2^j\delta \ge |x|$. For $j < j_0$ the minimum in (11) is $|x|^{-d}$ and the terms are $\varepsilon(2^j\delta)^{-1}|x|^{-d}$, a geometric series in 2^{-j} summing to $\lesssim \varepsilon\delta^{-1}|x|^{-d}$ (dominated by $j=0$). For $j \ge j_0$ the minimum is $(2^j\delta)^{-d}$ and the terms are $\varepsilon(2^j\delta)^{-(d+1)}$, again geometric, summing to $\lesssim \varepsilon(2^{j_0}\delta)^{-(d+1)} \le \varepsilon\delta^{-1}|x|^{-d}$ whenever $|x| \ge \delta$. For $|x| < \delta$ only $j \ge j_0 = 0$ occurs and the sum is $\lesssim \varepsilon\delta^{-(d+1)} = \varepsilon\delta^{-1}\delta^{-d}$. Both cases are $(*)$.

2.5 The main integral. Then for small p and appropriate η, using $\partial_n F_\varepsilon = \partial_n G_\varepsilon - \partial_n \rho_\varepsilon$,

$$\begin{aligned}\int\Big(\big|1+\partial_n F_\varepsilon\big|^p - 1\Big) &= \underbrace{\int\Big(\big|1+\partial_n G_\varepsilon\big|^p - 1\Big)}_{<-2\eta \text{ by } (7)} + \int\Big(\big|1+\partial_n F_\varepsilon\big|^p - \big|1+\partial_n G_\varepsilon\big|^p\Big) \\ &\le -2\eta + \underbrace{\int_{|x|<1}\Big|\big|1+\partial_n F_\varepsilon\big|^p - \big|1+\partial_n G_\varepsilon\big|^p\Big|}_{=:\mathrm{I}} \\ &\qquad + \underbrace{\int_{|x|>1}\Big(\big|1+\partial_n F_\varepsilon\big|^p - \big|1+\partial_n G_\varepsilon\big|^p\Big)}_{=:\mathrm{II}}. \end{aligned} \tag{12}$$

The term I. By Background ,

$$\mathrm{I} \le \int_{|x|\le 1} \big|\partial_n \rho_\varepsilon\big|^p,$$

and $(*)$ implies this goes to 0 with ε.

EXPANSION — a step he compressed, written out

It goes to 0 — and this is where p is constrained. [B20] By $(*)$, $|\partial_n \rho_\varepsilon| \lesssim \varepsilon\delta^{-1}\min(\delta^{-d}, |x|^{-d})$, and $\varepsilon\delta^{-1} = \varepsilon^{1/2}$ since $\delta = \varepsilon^{1/2}$. Split at $|x| = \delta$.

- $|x| < \delta$: the integrand is $\lesssim (\varepsilon^{1/2}\delta^{-d})^p$ over a set of measure $\lesssim \delta^{d-1}$, giving $\varepsilon^{p/2}\delta^{-dp+d-1} = \varepsilon^{\frac{p}{2}}\varepsilon^{\frac{-dp+d-1}{2}} = \varepsilon^{\frac{(d-1)(1-p)}{2}} \longrightarrow 0$ for $p < 1$.
- $\delta < |x| \leq 1$: the integrand is $\lesssim \varepsilon^{p/2}|x|^{-dp}$, and
$$\int_\delta^1 r^{-dp}\, r^{d-2}\, dr = \int_\delta^1 r^{\,d-2-dp}\, dr,$$
which stays bounded as $\delta \downarrow 0$ *iff* $d-2-dp > -1$, i.e.
$$\boxed{p < \tfrac{d-1}{d}}$$
and then the whole piece is $\lesssim \varepsilon^{p/2} \to 0$.

The paper says only "for small p". The explicit constraint is $p < \frac{d-1}{d}$, which for $d \geq 3$ means $p < \frac{2}{3}$; combined with $p < 1$ from Background the binding constraint is $p < \frac{d-1}{d}$. Together with whatever "below a certain critical number" means in (7), this fixes $p_0(d)$.

The term II. When $|x| > 1$ we have a lower bound for $|1 + \partial_n G_\varepsilon|$, and $|\partial_n \rho_\varepsilon|$ is small; it follows by the mean value theorem that $\big||1 + \partial_n F_\varepsilon|^p - |1 + \partial_n G_\varepsilon|^p\big| \leq C|\partial_n \rho_\varepsilon|$, and therefore

$$\mathrm{II} \leq C \int_{|x|>1} |\partial_n \rho_\varepsilon| \leq C\varepsilon\delta^{-p},$$

which again goes to 0 with ε. That proves (3).

2.6 Two repairs in the term II.

CAUTION — not fully justified here; OPEN-GAP in the ledger

[downgraded: repaired below] **The lower bound fails exactly at** $|x| = 1$**.** [B21] By (6), in the far field $\partial_n G_\varepsilon \approx -|x|^{-d}$, so
$$1 + \partial_n G_\varepsilon \approx 1 - |x|^{-d},$$
which *vanishes at* $|x| \approx 1$ and is only $\approx d(|x| - 1)$ just beyond it. So "we have a lower bound for $|1 + \partial_n G_\varepsilon|$ when $|x| > 1$" is not true as stated: the mean value theorem is being applied across the very point where the function it is applied to has a zero. The following repair costs nothing.

FILLED — an argument omitted in the original, supplied here

Repair: split $\{|x| > 1\}$ at $|x| = 2$. *(a) On* $\{1 < |x| < 2\}$, do not use the mean value theorem. Background gives directly
$$\int_{1<|x|<2} \Big| |1 + \partial_n F_\varepsilon|^p - |1 + \partial_n G_\varepsilon|^p \Big| \leq \int_{1<|x|<2} |\partial_n \rho_\varepsilon|^p \lesssim (\varepsilon^{1/2})^p \longrightarrow 0,$$
using $|\partial_n \rho_\varepsilon| \lesssim \varepsilon^{1/2}|x|^{-d} \leq \varepsilon^{1/2}$ there and the finite measure of the annulus. This is the same computation as for term I, on a region of measure $O(1)$.
(b) On $\{|x| > 2\}$ the lower bound is genuine: $|x|^{-d} \leq 2^{-d} \leq \frac{1}{2}$, so $1 + \partial_n G_\varepsilon \geq 1 - 2^{-d} \geq \frac{1}{2}$; and $|\partial_n \rho_\varepsilon| \lesssim \varepsilon^{1/2}|x|^{-d} \leq \varepsilon^{1/2}2^{-d}$, so also $1 + \partial_n F_\varepsilon \geq \frac{1}{2} - \varepsilon^{1/2} \geq \frac{1}{4}$ for ε small. Background

with $c = \frac{1}{4}$ then gives $\big||1+\partial_n F_\varepsilon|^p - |1+\partial_n G_\varepsilon|^p\big| \le p4^{1-p}\,|\partial_n\rho_\varepsilon|$, so

$$\int_{|x|>2} |\cdots| \le C\int_{|x|>2} |\partial_n\rho_\varepsilon|.$$

The repaired split changes nothing downstream: both pieces $\to 0$ with ε, which is all (12) needs.

EXPANSION — a step he compressed, written out

The exponent in $C\varepsilon\delta^{-p}$. [B23] What $(*)$ yields is

$$\int_{|x|>2} |\partial_n\rho_\varepsilon| \lesssim \varepsilon\delta^{-1}\int_{|x|>2} |x|^{-d}dx = c\,\varepsilon\delta^{-1} = c\,\varepsilon^{1/2},$$

since $\int_{|x|>2}|x|^{-d}dx = c\int_2^\infty r^{-d}r^{d-2}dr = c\int_2^\infty r^{-2}dr < \infty$ in $\mathbb{R}^{d-1}$. The original prints the exponent $-p$ rather than -1; as $\delta < 1$ and $p < 1$, $\delta^{-p} < \delta^{-1}$, so the printed bound is the *stronger* of the two and is not what $(*)$ delivers. Since both $\varepsilon\delta^{-1} = \varepsilon^{1/2}$ and $\varepsilon\delta^{-p} = \varepsilon^{1-p/2}$ tend to 0, the conclusion is unaffected. We use $\varepsilon^{1/2}$.

2.7 The gradient bound (4)**.** To prove $|\nabla\widehat{F}_\varepsilon| \lesssim \min(\varepsilon^{-d}, |x|^{-d})$: this estimate is obvious for $|\nabla G_\varepsilon|$ and for $|\nabla_T\rho_\varepsilon|$; for $|\partial_n\rho_\varepsilon|$ it follows from $(*)$.

EXPANSION — a step he compressed, written out

"Obvious", done. [B24] Since $\nabla\widehat{F}_\varepsilon = \nabla G_\varepsilon - \nabla\rho_\varepsilon$ and $\nabla = (\nabla_T, \partial_n)$, there are three pieces.
(i) ∇G_ε. The normal component is (6): $|\partial_n G_\varepsilon| = \varepsilon^{-d}|h(s)|$ with h bounded, so $|\partial_n G_\varepsilon| \le (d-1)\varepsilon^{-d}$ always, while for $|x| \gg \varepsilon$, $|h(s)| \approx s^{-d}$ gives $|\partial_n G_\varepsilon| \approx |x|^{-d}$. Hence $|\partial_n G_\varepsilon| \lesssim \min(\varepsilon^{-d}, |x|^{-d})$ — this is exactly the shape asserted, and it is the term that forces the ε^{-d} in (4). The tangential component is $|\nabla_T G_\varepsilon| \lesssim \varepsilon|x|^{-(d+1)}$ from §2.4, which is smaller.
(ii) $\nabla_T\rho_\varepsilon$. On $\operatorname{supp}\psi_j$, §2.4 gives $|\nabla_T\rho_\varepsilon^j| \lesssim \varepsilon(2^j\delta)^{-(d+1)} \approx \varepsilon|x|^{-(d+1)}$; the ψ_j have bounded overlap, so $|\nabla_T\rho_\varepsilon| \lesssim \varepsilon|x|^{-(d+1)}$, and since ρ_ε is supported in $\{|x| > \delta/2\}$ this is $\le 2\varepsilon\delta^{-1}|x|^{-d} = 2\varepsilon^{1/2}|x|^{-d} \le |x|^{-d}$.
(iii) $\partial_n\rho_\varepsilon$. By $(*)$, $|\partial_n\rho_\varepsilon| \lesssim \varepsilon^{1/2}\min(\delta^{-d}, |x|^{-d})$. The second entry is already $\le |x|^{-d}$; for the first, $\varepsilon^{1/2}\delta^{-d} = \varepsilon^{\frac{1-d}{2}} \le \varepsilon^{-d}$ because $\frac{d-1}{2} \le d$ for $d \ge 1$.
Adding, $|\nabla\widehat{F}_\varepsilon| \lesssim \min(\varepsilon^{-d}, |x|^{-d})$. $\square$

§3. Lemma 2: the correction step on a cube

Lemma 2. *If N is large enough there is a constant $\beta = \beta(N) > 0$ such that: if $Q \subset \mathbb{R}^{d-1}$ is a cube, a_Q its centre and $I : Q \to \mathbb{R}$ a function such that*

$$N^{d-1}\,|I(a_Q)|^{-1}\sup_{x\in Q}|I(x) - I(a_Q)| \tag{13}$$

is sufficiently small (independently of N and ε), then

$$\Big(\int_Q \Big|I(x) + \partial_n F_\varepsilon\big(N\ell(Q)^{-1}(x-a_Q)\big)\Big|^p dx\Big)^{1/p} < e^{-2\beta}\,|I(a_Q)|\,|Q|^{1/p}. \tag{14}$$

OUR NOTATION — not in the original

The original prints $\beta \subset \beta(N)$ for $\beta = \beta(N)$. [C1]

FILLED — an argument omitted in the original, supplied here

Order of quantifiers. [C2] Read: there are $N_0(d,p)$ and an absolute threshold $\tau = \tau(d,p) > 0$ such that for every $N \geq N_0$ there is $\beta = \beta(N,d,p) > 0$ such that for all $\varepsilon < \varepsilon_0(p,d)$ (as in Lemma 1), all cubes Q, and all I with (13) $< \tau$, the conclusion (14) holds. The parenthetical "independently of N and ε" refers to τ, not to β: β certainly depends on N.

3.1 Reduction to $Q = Q(N)$ and $I(a_Q) = 1$. We can assume $Q = Q(N)$. It is then equivalent to show that if $N^{d-1}\|I-1\|_\infty$ is sufficiently small then

$$\int_{Q(N)} \Big(|I + \partial_n F_\varepsilon|^p - 1 \Big) \leq -\eta \tag{15}$$

for suitable $\eta > 0$.

EXPANSION — a step he compressed, written out

The two reductions. [C3] *Scaling.* The map $\Phi(x) = N\ell(Q)^{-1}(x - a_Q)$ carries Q onto $Q(N)$ and a_Q to 0; it has constant Jacobian $(N/\ell(Q))^{d-1}$, and $|Q| = \ell(Q)^{d-1}$, $|Q(N)| = N^{d-1}$, so $|Q(N)|/|Q| = (N/\ell(Q))^{d-1}$. Writing $\tilde{I} = I \circ \Phi^{-1}$ and substituting $y = \Phi(x)$,

$$\int_Q \big|I(x) + \partial_n F_\varepsilon(\Phi(x))\big|^p dx = \frac{|Q|}{|Q(N)|} \int_{Q(N)} \big|\tilde{I}(y) + \partial_n F_\varepsilon(y)\big|^p dy.$$

Since $|Q|^{1/p}$ appears on the right of (14), the inequality is equivalent to $\frac{1}{|Q(N)|}\int_{Q(N)} |\tilde{I} + \partial_n F_\varepsilon|^p < e^{-2\beta p}|\tilde{I}(0)|^p$, which involves Q only through $\tilde{I}$; and (13) is invariant under Φ because it is built from the values of I alone.

Normalisation. Both sides of the resulting inequality are homogeneous of degree p in $(\tilde{I}, \partial_n F_\varepsilon)$ jointly — but $\partial_n F_\varepsilon$ is fixed, so we cannot rescale it. What we can do is divide by $|\tilde{I}(0)|$: it is enough to prove the statement when $\tilde{I}(0) = 1$, replacing F_ε by $F_\varepsilon/|\tilde{I}(0)|$ — and here the argument uses that Lemma 1 is applied to a *fixed* F_ε while I varies. In the application (§6) I is $\partial_n u_n(a_Q)$ plus a small perturbation, and the factor $\partial_n u_n(a_Q)$ is exactly the coefficient multiplying $F_{\varepsilon_{n+1}}$ in the definition of u_{n+1}, so the two scalings match. With $\tilde{I}(0) = 1$, (13) becomes $N^{d-1}\|\tilde{I} - 1\|_\infty$, and (14) becomes (15) with $e^{-2\beta p} = 1 - \eta/N^{d-1}$; see §3.4.

3.2 The estimate. This last inequality is true since

$$\begin{aligned}\int_{Q(N)} \Big(|I + \partial_n F_\varepsilon|^p - |1 + \partial_n F_\varepsilon|^p \Big) dx &\leq \int_{Q(N)\setminus D(0,1)} + \int_{D(0,1)} \\ &\leq C\int_{Q(N)\setminus D(0,1)} |I-1|\, dx + \int_{D(0,1)} |I-1|^p dx \end{aligned} \tag{16}$$

(because $1 + \partial_n F_\varepsilon$ is bounded away from 0 when $|x| > 1$)

$$\leq C\|I-1\|_\infty N^{d-1} + C\|I-1\|_\infty^p,$$

which is small. It remains to apply Lemma 1.

EXPANSION — a step he compressed, written out

Three corrections in that chain.

- [C4] The original prints both terms of the first line as $|1+\partial_n F_\varepsilon|^p$, so the left side is identically 0. The first must be $|I + \partial_n F_\varepsilon|^p$: the whole point is to compare the I-version with the 1-version, which is what Lemma 1 controls. We have restored I.

- [C6] The original's last line prints $C\|I-1\|_\infty$ for the second term. That term is $\int_{D(0,1)}|I-1|^p \le |D(0,1)|\cdot\|I-1\|_\infty^p$, so the exponent is p, not 1. Since $\|I-1\|_\infty < 1$ and $p<1$ we have $\|I-1\|_\infty^p > \|I-1\|_\infty$, so the printed bound is the smaller one and is not what the line before yields. Harmless: the hypothesis $N^{d-1}\|I-1\|_\infty < \tau$ with $N \ge 1$ forces $\|I-1\|_\infty < \tau$, hence $\|I-1\|_\infty^p < \tau^p$, and τ is at our disposal.
- The first term: $\int_{Q(N)\setminus D(0,1)}|I-1| \le \|I-1\|_\infty |Q(N)| = \|I-1\|_\infty N^{d-1}$, which is where the N^{d-1} in (13) comes from — and it is the only place. This is why the hypothesis carries the factor N^{d-1} and not, say, N^d.

3.3 The same lower-bound repair.

CAUTION — not fully justified here; OPEN-GAP in the ledger

[downgraded: repaired below] The parenthetical "$1+\partial_n F_\varepsilon$ is bounded away from 0 when $|x|>1$" has exactly the defect of §2.6: by (6) and §2.4, $\partial_n F_\varepsilon = \partial_n G_\varepsilon - \partial_n \rho_\varepsilon \approx -|x|^{-d}$ in the far field, so $1+\partial_n F_\varepsilon$ has a zero near $|x|=1$. [C5]

FILLED — an argument omitted in the original, supplied here

Repair. Split $Q(N)\backslash D(0,1)$ into the annulus $A=\{1<|x|<2\}$ and $Q(N)\backslash D(0,2)$. On A, use Background : the integrand is $\le |I-1|^p$, so this piece contributes $\le |A|\cdot\|I-1\|_\infty^p \le C\|I-1\|_\infty^p$ — the same shape as the $D(0,1)$ term, and absorbed into it. On $Q(N)\setminus D(0,2)$ the bound $|1+\partial_n F_\varepsilon| \ge \frac{1}{4}$ of §2.6(b) holds, so Background applies and gives $C|I-1|$ pointwise, hence $C\|I-1\|_\infty N^{d-1}$. The final bound is therefore $C\|I-1\|_\infty N^{d-1} + C\|I-1\|_\infty^p$, exactly as in §3.2, and the argument proceeds unchanged.

3.4 Applying Lemma 1, and what β is.

FILLED — an argument omitted in the original, supplied here

"It remains to apply lemma 1" — the tail. [C7] Lemma 1 controls $\int_{\mathbb{R}^{d-1}}(|1+\partial_n F_\varepsilon|^p - 1) < -\eta_1$, an integral over the whole hyperplane, whereas (15) is over $Q(N)$. The difference is

$$\int_{\mathbb{R}^{d-1}\backslash Q(N)} \Big(\big|1+\partial_n F_\varepsilon\big|^p - 1\Big),$$

and $\mathbb{R}^{d-1}\setminus Q(N) \subset \{|x|>N/2\}$. There $|\partial_n F_\varepsilon| \lesssim |x|^{-d} \le (N/2)^{-d} \le \frac{1}{2}$ for N large, so by Background with $c=\frac{1}{2}$, $\big||1+\partial_n F_\varepsilon|^p - 1\big| \le p2^{1-p}|\partial_n F_\varepsilon| \lesssim |x|^{-d}$, and

$$\Big|\int_{\mathbb{R}^{d-1}\backslash Q(N)}\Big| \lesssim \int_{|x|>N/2}|x|^{-d}dx = c\,N^{-1}.$$

Choosing N large enough that $cN^{-1} < \eta_1/2$ gives

$$\int_{Q(N)}\Big(\big|1+\partial_n F_\varepsilon\big|^p - 1\Big) < -\tfrac{\eta_1}{2}.$$

Adding (16), whose right-hand side is $<\eta_1/4$ once $N^{d-1}\|I-1\|_\infty < \tau$ with τ small, yields (15) with $\eta = \eta_1/4$. *Note the order*: η_1 comes from Lemma 1 and depends on p, d only; N is then chosen; then τ; and only then β.

EXPANSION — a step he compressed, written out

Where $e^{-2\beta}$ comes from, and what it depends on. [C8] (15) says $\int_{Q(N)} |I + \partial_n F_\varepsilon|^p \leq |Q(N)| - \eta$. Hence

$$\Big(\int_{Q(N)} \big|I + \partial_n F_\varepsilon\big|^p\Big)^{1/p} \leq \big(1 - \eta/N^{d-1}\big)^{1/p} |Q(N)|^{1/p},$$

so (14) holds with

$$\boxed{\beta = \beta(N) = -\frac{1}{2p}\log\big(1 - \eta N^{-(d-1)}\big) > 0}$$

which depends on p, d and N — and on nothing else. This is the β that governs the geometric decay of the whole construction, and it is fixed once and for all as soon as N is. □

§4. The recursive construction

For a sufficiently large N to be determined, choose β according to Lemma 2. Also fix sequences $K_n \to \infty$, $\varepsilon_n \searrow 0$ (to be determined). Constants denoted C will be independent of these choices. Let u_0 be a smooth function on $\mathbb{R}^{d-1}$ which vanishes on $Q(1)$.

FILLED — an argument omitted in the original, supplied here

u_0 vanishes *on* $Q(1)$ — this is correct, and is the single most confusing line in the paper. [D2] u_0 is *boundary data.* That $u_0 \equiv 0$ on $Q(1)$ does *not* make $\partial_n u_0$ vanish there: the normal derivative of the harmonic extension at a boundary point is determined by u_0 everywhere, not just nearby, and is in general nonzero on $Q(1)$. It is precisely this nonzero normal derivative that the recursion drives to 0. And, per §1, take $u_0 \not\equiv 0$, so that $f \not\equiv 0$.

OUR NOTATION — not in the original

The order in which everything is chosen [D4] — the paper gives only part of this, and the argument is unreadable without it:

$$d,\ p \longrightarrow \eta_1 \text{ (Lemma 1)} \longrightarrow N \longrightarrow \tau,\ \beta = \beta(N)$$
$$\longrightarrow A \longrightarrow \{K_n\}, \{\varepsilon_n\} \longrightarrow \delta_{n+1} \text{ at each stage.}$$

A constant written C may depend on d, p, N but never on $A, K_n, \varepsilon_n, \delta_n$ or n. Crucially δ_{n+1} is chosen *after* u_n exists, so it may depend on the modulus of continuity of $\partial_n u_n$ — used at D10 and D18 below.

We drop the $\widehat{\ }$ and identify functions on $\mathbb{R}^{d-1}$ with their harmonic extensions.

4.1 The inductive data. If the nth stage has been done ($n \geq 0$) we have: $\delta_n > 0$ with $\delta_n^{-1} \in \mathbb{Z}$; a subset $\mathcal{Q}_n \subset \mathcal{H}_n$, where $\mathcal{H}_n$ is the collection of $\delta_n^{-(d-1)}$ cubes of side δ_n whose union is $Q(1)$; and a smooth u_n with

$$\Big(\int_{V_n} \big|\partial_n u_n\big|^p\Big)^{1/p} \leq A\, e^{-\beta n}, \qquad V_n = \bigcup\{Q : Q \in \mathcal{Q}_n\}, \tag{17}$$

A a large constant independent of n. To start, $\delta_0 = 1$, $\mathcal{Q}_0 = \{Q(1)\}$.

EXPANSION — a step he compressed, written out

Base case. [D3] At $n = 0$, $V_0 = Q(1)$ and (17) reads $(\int_{Q(1)} |\partial_n u_0|^p)^{1/p} \leq A$. Since u_0 is smooth and $Q(1)$ has measure 1, $\partial_n u_0$ is bounded on $Q(1)$ and the left side is finite; take

$A \geq \|\partial_n u_0\|_{L^\infty(Q(1))}$. This is one of the constraints on A; §6 imposes the other, and both are lower bounds, so a single large A serves.

4.2 Stage $n+1$. Choose δ_{n+1} with $\delta_n/\delta_{n+1} \in \mathbb{Z}$, extremely small (how small is determined later). Then $\mathcal{Q}_{n+1}$ is all cubes $Q \in \mathcal{H}_{n+1}$ such that

(a) the cube $Q' \in \mathcal{H}_n$ with $Q \subset Q'$ satisfies $Q' \in \mathcal{Q}_n$, and
(b) $\left(\frac{1}{|Q|}\int_Q |\partial_n u_n|^p\right)^{1/p} < K_{n+1}e^{-\beta n}$.

Let a_Q be the centre of Q and define

$$u_{n+1}(x) = u_n(x) + \sum_{Q\in\mathcal{Q}_{n+1}} \partial_n u_n(a_Q)\, F_{\varepsilon_{n+1}}\big(N\delta_{n+1}^{-1}(x-a_Q)\big)\cdot\frac{\delta_{n+1}}{N}, \tag{18}$$

so that

$$\nabla u_{n+1}(x) = \nabla u_n(x) + \sum_{Q\in\mathcal{Q}_{n+1}} \partial_n u_n(a_Q)\, \nabla F_{\varepsilon_{n+1}}\big(N\delta_{n+1}^{-1}(x-a_Q)\big). \tag{19}$$

EXPANSION — a step he compressed, written out

Why the factor δ_{n+1}/N disappears. [D5] Writing $\Phi_Q(x) = N\delta_{n+1}^{-1}(x-a_Q)$, the chain rule gives $\nabla\big[F(\Phi_Q(x))\big] = N\delta_{n+1}^{-1}(\nabla F)(\Phi_Q(x))$, and the prefactor δ_{n+1}/N in (18) cancels it exactly. This is the reason for that factor: the *amplitude* of the correction is $O(\delta_{n+1}/N)$, so u barely moves, while its *gradient* is $O(1)$ — which is what has to be $O(1)$ for Lemma 2 to bite. Without the factor the corrections would not converge in C^0; with a larger factor they would not correct the gradient at all.

4.3 Support of the correction.

EXPANSION — a step he compressed, written out

[D6, D29] supp $F_\varepsilon \subset D(0,\varepsilon^{1/2})$, so the Q-term of (18) is supported where $|N\delta_{n+1}^{-1}(x-a_Q)| < \varepsilon_{n+1}^{1/2}$, i.e. in

$$D\Big(a_Q,\ \frac{\delta_{n+1}\varepsilon_{n+1}^{1/2}}{N}\Big).$$

The original writes the radius as $\delta_{n+1}\varepsilon_{n+1}^{1/2}$, omitting the $1/N$; since N is fixed this only changes constants, and we keep the correct radius. For δ_{n+1} small this disc is contained in Q, hence in $Q(1)$. Consequently

$$u_{n+1} = u_n \ \text{ on } \mathbb{R}^{d-1}\setminus Q(1) \quad \text{for every } n, \qquad \text{so} \qquad f = u_0 \ \text{ on } \mathbb{R}^{d-1}\setminus Q(1),$$

which is the fact §1 needs for $f \not\equiv 0$.

§5. Lemma 3 and its corollary: controlling the interference

Lemma 3. *For all $\rho >$ const $\cdot\, \delta_{n+1}$ and $x \in \mathbb{R}^{d-1}$,*

$$\sum_{\substack{Q\in\mathcal{Q}_{n+1}\\ |x-a_Q|>\rho}} \big|\nabla F_{\varepsilon_{n+1}}\big(N\delta_{n+1}^{-1}(x-a_Q)\big)\big| \leq C\, N^{-d}\,\delta_{n+1}\,\rho^{-1}, \tag{20}$$

where C depends on a lower bound for ρ/δ_{n+1}.

Proof. The left hand side is $\leq \sum_{|x-a_Q|>\rho}\big(N\delta_{n+1}^{-1}|x-a_Q|\big)^{-d}$ by the last part of Lemma 1. Lemma 3 follows by considering $\sum \delta_{n+1}^{d-1}|x-a_Q|^{-d}$ as a Riemann sum for $\int_{|x|>\rho}|x|^{-d}$, which is justified as long as ρ/δ_{n+1} does not approach 0. $\square$

EXPANSION — a step he compressed, written out

The arithmetic. [D7] By (4) applied to $\varepsilon = \varepsilon_{n+1}$ at the point $y = N\delta_{n+1}^{-1}(x - a_Q)$, $|\nabla F_{\varepsilon_{n+1}}(y)| \lesssim |y|^{-d} = N^{-d}\delta_{n+1}^{d}|x - a_Q|^{-d}$; the other entry ε^{-d} of the minimum is not used here because $|x - a_Q| > \rho$ keeps $|y|$ large. Pulling out the constants,

$$\text{LHS} \;\lesssim\; N^{-d}\delta_{n+1}^{d} \sum_{|x-a_Q|>\rho} |x - a_Q|^{-d} \;=\; N^{-d}\,\delta_{n+1} \cdot \Big[\delta_{n+1}^{d-1} \sum_{|x-a_Q|>\rho} |x - a_Q|^{-d}\Big].$$

The bracket is a Riemann sum: the centres a_Q of the cubes of $\mathcal{H}_{n+1}$ form a lattice of spacing δ_{n+1} in $\mathbb{R}^{d-1}$, so each contributes a cell of volume δ_{n+1}^{d-1}, and the bracket approximates

$$\int_{|y|>\rho} |y|^{-d}\,dy = \omega_{d-2}\int_{\rho}^{\infty} r^{-d} r^{d-2}\,dr = \omega_{d-2}\int_{\rho}^{\infty} r^{-2}\,dr = \omega_{d-2}\,\rho^{-1}$$

in $\mathbb{R}^{d-1}$. Combining gives (20). Note where $d \geq 2$ is needed for the integral to converge at infinity — it always does, since the exponent is -2 independently of d.

FILLED — an argument omitted in the original, supplied here

The Riemann-sum comparison, made quantitative. [D8] Let $L = \rho/\delta_{n+1}$, assumed $\geq L_0$ for some fixed $L_0 > 1$ — this is the "$\rho > \text{const} \cdot \delta_{n+1}$" and the "$C$ depends on a lower bound for ρ/δ_{n+1}" of the statement. For a lattice point a_Q with $|x - a_Q| > \rho$, every y in its cell $a_Q + [-\frac{\delta_{n+1}}{2}, \frac{\delta_{n+1}}{2}]^{d-1}$ satisfies $|x - y| \geq |x - a_Q| - \frac{\sqrt{d-1}}{2}\delta_{n+1} \geq |x - a_Q|\big(1 - \frac{\sqrt{d-1}}{2L}\big)$. Choosing $L_0 = \sqrt{d-1}$ gives $|x - y| \geq \frac{1}{2}|x - a_Q|$, hence $|x - a_Q|^{-d} \leq 2^d|x - y|^{-d}$ and

$$\delta_{n+1}^{d-1} \sum_{|x-a_Q|>\rho} |x - a_Q|^{-d} \leq 2^d \int_{|x-y|>\rho/2} |x - y|^{-d} dy = 2^d\,\omega_{d-2}\,(\rho/2)^{-1},$$

the cells being disjoint and contained in $\{|x - y| > \rho/2\}$. So $C = 2^{d+1}\omega_{d-2}$ works, and the dependence on the lower bound L_0 is explicit. Without $L \geq L_0$ the cell containing x itself could be included and the sum would diverge — which is exactly the caveat the paper states.

Corollary. *If δ_{n+1} is small enough, then on $\mathbb{R}^{d-1}$,*

$$|\nabla u_{n+1} - \nabla u_n| \leq C\,K_{n+1}\,e^{-\beta n}\,\varepsilon_{n+1}^{-d}. \tag{21}$$

Proof. Assume for notational purposes $x \in Q \in \mathcal{Q}_{n+1}$ — the other case is a little simpler. If δ_{n+1} is small, then by (b) and smoothness of u_n,

$$\big|\partial_n u_n(a_{Q'})\big| \leq 2K_{n+1}e^{-\beta n} \qquad \text{for all } Q' \in \mathcal{Q}_{n+1}. \tag{22}$$

So, by (19),

$$\begin{aligned} |\nabla u_{n+1}(x) - \nabla u_n(x)| \leq &\sum_{\substack{Q' \in \mathcal{Q}_{n+1} \\ Q' \neq Q}} \big|\partial_n u_n(a_{Q'})\big|\,\big|\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'}))\big| \\ &+ \big|\partial_n u_n(a_Q)\big|\,\big|\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_Q))\big| \\ \leq &\; 2K_{n+1}e^{-\beta n}\big(C + C\varepsilon_{n+1}^{-d}\big), \end{aligned}$$

the sum by Lemma 3 and the remaining term by Lemma 1. $\square$

EXPANSION — a step he compressed, written out

Two things to fix and one to justify.

- [D9] The original writes $|\nabla u_n(a_{Q'})|$ throughout this proof. But the coefficient in (19) is $\partial_n u_n(a_{Q'})$, the *normal* derivative, and hypothesis (b) controls only the normal derivative. We have written $\partial_n u_n(a_{Q'})$. (Since $|\partial_n u_n| \le |\nabla u_n|$, the printed version is an over-estimate of the coefficient, but it is not one that (b) supports.)
- [D10] (22) is the step "by (b) and smoothness of u_n". Hypothesis (b) is an L^p average over Q'; to convert it to the pointwise value at $a_{Q'}$, note that $\partial_n u_n$ is continuous, so $|\partial_n u_n(a_{Q'})|^p \le \frac{1}{|Q'|}\int_{Q'} |\partial_n u_n|^p + \omega(\delta_{n+1})$ where ω is a modulus of continuity for $|\partial_n u_n|^p$ on $Q(1)$. Choosing δ_{n+1} small enough that $\omega(\delta_{n+1}) \le (2^p-1)K_{n+1}^p e^{-p\beta n}$ gives $|\partial_n u_n(a_{Q'})| \le 2K_{n+1}e^{-\beta n}$. This is legitimate because δ_{n+1} is chosen after u_n (see the ordering block in §4); ω depends on n, and that is allowed.
- [D11] The two terms. For $Q' \ne Q$ apply Lemma 3 with $\rho = \frac{1}{2}\delta_{n+1}$: distinct cube centres are $\ge \delta_{n+1}$ apart, so every $Q' \ne Q$ has $|x - a_{Q'}| \ge \frac{1}{2}\delta_{n+1}$ when $x \in Q$; the sum is then $\le CN^{-d}\delta_{n+1}(\frac{1}{2}\delta_{n+1})^{-1} = 2CN^{-d} \le C$. For $Q' = Q$ we cannot use Lemma 3 (x may equal a_Q), and instead use the other entry of the minimum in (4): $|\nabla F_{\varepsilon_{n+1}}| \lesssim \varepsilon_{n+1}^{-d}$. That single term is the sole source of the factor ε_{n+1}^{-d} in (21), and hence the sole reason condition (1) of §7 carries ε_{n+1}^{-d}.

§6. Lemma 4: the induction step

Lemma 4. *If A is large then* $\left(\int_{V_{n+1}} \left|\partial_n u_{n+1}\right|^p\right)^{1/p} < A\, e^{-\beta(n+1)}$.

6.1 The interference estimate $(*)$**.** First we claim: given $\gamma > 0$, if δ_{n+1} is small enough then for any $x \in Q \in \mathcal{Q}_{n+1}$

$$\sum_{\substack{Q' \in \mathcal{Q}_{n+1} \\ Q' \ne Q}} \left|\partial_n u_n(a_{Q'})\right| \left|\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'}))\right| \le C\, N^{-d}\left|\partial_n u_n(x)\right| + \gamma. \tag{$*$}$$

FILLED — an argument omitted in the original, supplied here

Order of choices in $(*)$**.** [D14] The printed proof "fixes" $M < \infty$, then at the end says M is arbitrary. Read it as: given $\gamma > 0$, first choose M so large that the far part is $< \gamma/2$; then choose δ_{n+1} so small that the near part is within $\gamma/2$ of its limit. Both choices are made after u_n, K_{n+1}, ε_{n+1} are known.

Proof of $(*)$. Fix $M < \infty$ and split the sum at $|x - a_{Q'}| = M\delta_{n+1}$.

Near part.

$$\begin{aligned}
\sum_{|x-a_{Q'}|<M\delta_{n+1}} &\le \sup\Big\{\left|\partial_n u_n(a_{Q'})\right| : |x - a_{Q'}| < M\delta_{n+1}\Big\} \cdot \sum_{Q' \ne Q} \left|\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'}))\right| \\
&< CN^{-d} \sup\Big\{\left|\partial_n u_n(a_{Q'})\right| : |x - a_{Q'}| < M\delta_{n+1}\Big\} \qquad (\text{Lemma 3 with } \rho = \tfrac{1}{2}\delta_{n+1}) \\
&< CN^{-d}\Big(\left|\partial_n u_n(x)\right| + \gamma\Big),
\end{aligned}$$

where $\gamma \to 0$ as $\delta_{n+1} \to 0$ by continuity of $|\partial_n u_n|$.

Far part.

$$\begin{aligned}\sum_{|x-a_{Q'}|>M\delta_{n+1}} &\leq \sup\bigl\{|\partial_n u_n(y)| : y \in V_{n+1}\bigr\}\cdot \sum_{|x-a_{Q'}|>M\delta_{n+1}} \bigl|\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x-a_{Q'}))\bigr| \\ &\leq 2K_{n+1}e^{-\beta n}\sum_{|x-a_{Q'}|>M\delta_{n+1}} \bigl|\nabla F_{\varepsilon_{n+1}}(\cdots)\bigr| \qquad (\delta_{n+1}\text{ small enough}) \\ &\leq C\,K_{n+1}\,e^{-\beta n}\,M^{-1} \qquad (\text{Lemma 3 with } \rho = M\delta_{n+1}).\end{aligned}$$

Since M is arbitrary we can make this as small as we want, which proves $(*)$. □

EXPANSION — a step he compressed, written out

[D12, D13] Two details. The near part uses Lemma 3 with $\rho = \frac{1}{2}\delta_{n+1}$, giving $CN^{-d}\delta_{n+1}\cdot 2\delta_{n+1}^{-1} = 2CN^{-d}$; the sup is then handled by continuity, as in D10. The original writes "$\gamma \to 0$ as $\delta_n \to 0$"; it is δ_{n+1}, the side of the stage-$(n+1)$ cubes, that shrinks and forces $a_{Q'} \to x$. The far part uses Lemma 3 with $\rho = M\delta_{n+1}$, giving $CN^{-d}\delta_{n+1}(M\delta_{n+1})^{-1} = CN^{-d}M^{-1}$, and the pointwise bound (22) for the coefficients.

6.2 Type 1 and type 2 cubes. Call $Q \in \mathcal{Q}_{n+1}$ *type 1* if $|\partial_n u_n(a_Q)| > e^{-4\beta(n+1)}$, and *type 2* otherwise. Fix $Q \in \mathcal{Q}_{n+1}$ and write, for $x \in Q$,

$$\partial_n u_{n+1}(x) = \underbrace{\partial_n u_n(a_Q)}_{\text{the "}I(a_Q)\text{"}} + \mathrm{Br}(x) \;+\; \partial_n u_n(a_Q)\,\partial_n F_{\varepsilon_{n+1}}\bigl(N\delta_{n+1}^{-1}(x-a_Q)\bigr), \tag{23}$$

where the *bracket* is

$$\mathrm{Br}(x) = \partial_n u_n(x) - \partial_n u_n(a_Q) + \sum_{\substack{Q'\in\mathcal{Q}_{n+1}\\ Q'\neq Q}} \partial_n u_n(a_{Q'})\,\partial_n F_{\varepsilon_{n+1}}\bigl(N\delta_{n+1}^{-1}(x-a_{Q'})\bigr). \tag{24}$$

EXPANSION — a step he compressed, written out

[D16] The printed bracket is unbalanced (the closing bracket sits after the sum, and the opening one before $\partial_n u_n(x)$); (23) is the reading that makes it an identity, and it is an identity: taking ∂_n of (18) and adding and subtracting $\partial_n u_n(a_Q)$ gives exactly these three groups. The decomposition is engineered so that $I := \partial_n u_n(a_Q) +$ *bracket is nearly constant on* Q, which is what Lemma 2 requires, while the last term is the correction Lemma 2 acts on.

By $(*)$ and smoothness of $\partial_n u_n$, the bracket may be made less than $CN^{-d}|\partial_n u_n(a_Q)| + \gamma$ with γ arbitrarily small.

6.3 Type 1 cubes. For Q type 1 the bracket is therefore $< CN^{-d}|\partial_n u_n(a_Q)|$ provided γ is chosen right. So Lemma 2 applies (using here that $CN^{-d} < cN^{-(d-1)}$ for large N) and we get

$$\Bigl(\int_Q |\partial_n u_{n+1}|^p\Bigr)^{1/p} < e^{-2\beta}|\partial_n u_n(a_Q)||Q|^{1/p} < e^{-\frac{3}{2}\beta}\Bigl(\int_Q |\partial_n u_n|^p\Bigr)^{1/p} \quad \text{if } \delta_{n+1} \text{ is small.} \tag{25}$$

EXPANSION — a step he compressed, written out

Checking Lemma 2's hypothesis. [D17] Put $I(x) = \partial_n u_n(a_Q) + \mathrm{bracket}(x)$, so $I(a_Q) = \partial_n u_n(a_Q)$ (the bracket vanishes at $x = a_Q$: its first two terms cancel and the sum is evaluated at $x = a_Q$ — strictly, the sum does not vanish at a_Q, so $I(a_Q)$ differs from $\partial_n u_n(a_Q)$ by at most the same bound, which is absorbed). Then

$$N^{d-1}|I(a_Q)|^{-1}\sup_Q |I - I(a_Q)| \le N^{d-1}\cdot \frac{2CN^{-d}|\partial_n u_n(a_Q)|}{|\partial_n u_n(a_Q)|} = 2CN^{-1},$$

which is $< \tau$ for N large. *This is precisely the parenthetical remark* "we're using here that $CN^{-d} < cN^{-(d-1)}$ for large N": the gain of one power of N between the bracket's N^{-d} and the hypothesis's N^{d-1} is what makes the hypothesis verifiable, and it is why type 1 is defined by a lower bound on $|\partial_n u_n(a_Q)|$ — without one, dividing by $|I(a_Q)|$ is uncontrolled. The γ is absorbed because for type 1, $|\partial_n u_n(a_Q)| > e^{-4\beta(n+1)}$ is a fixed positive number at this stage, so γ may be taken far smaller.

The second inequality of (25)**.** [D18] We must pass from $|\partial_n u_n(a_Q)||Q|^{1/p}$ to $(\int_Q |\partial_n u_n|^p)^{1/p}$. By continuity of $\partial_n u_n$, for δ_{n+1} small, $\frac{1}{|Q|}\int_Q |\partial_n u_n|^p \ge |\partial_n u_n(a_Q)|^p - \omega(\delta_{n+1}) \ge e^{-\frac{1}{2}\beta p}|\partial_n u_n(a_Q)|^p$, the last step because $|\partial_n u_n(a_Q)| > e^{-4\beta(n+1)}$ is bounded below at this stage while $\omega(\delta_{n+1}) \to 0$. Taking p-th roots, $|\partial_n u_n(a_Q)||Q|^{1/p} \le e^{\frac{1}{2}\beta}(\int_Q |\partial_n u_n|^p)^{1/p}$, and $e^{-2\beta}\cdot e^{\frac{1}{2}\beta} = e^{-\frac{3}{2}\beta}$.

6.4 Type 2 cubes. If Q is type 2 the bracket is $< CN^{-d}|\partial_n u_n(a_Q)| + \gamma$, which may be taken $< e^{-4\beta(n+1)}$. So

$$\int_Q |\partial_n u_{n+1}|^p \le |\partial_n u_n(a_Q)|^p \int_Q \Big|1 + \partial_n F_{\varepsilon_{n+1}}\big(N\delta_{n+1}^{-1}(x-a_Q)\big)\Big|^p + \int_Q [\,\cdot\,]^p \le 2\,e^{-4\beta(n+1)p}\,|Q|, \quad (26)$$

since (say) the first integral is $< |Q|$ by Lemma 2.

EXPANSION — a step he compressed, written out

[D19] Unpacking: by (23) and Background , $|\partial_n u_{n+1}|^p \le |\partial_n u_n(a_Q)|^p|1 + \partial_n F_{\varepsilon_{n+1}}(\cdots)|^p + |\text{bracket}|^p$. For type 2, $|\partial_n u_n(a_Q)|^p \le e^{-4\beta(n+1)p}$ and the bracket is $< e^{-4\beta(n+1)}$, so the second integral is $\le e^{-4\beta(n+1)p}|Q|$. For the first, we need $\int_Q |1+\partial_n F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x-a_Q))|^p < |Q|$: by the scaling of §3.1 this equals $\frac{|Q|}{|Q(N)|}\int_{Q(N)} |1+\partial_n F_\varepsilon|^p$, and §3.4 showed $\int_{Q(N)}(|1+\partial_n F_\varepsilon|^p - 1) < -\eta_1/2 < 0$, i.e. $\int_{Q(N)} |1+\partial_n F_\varepsilon|^p < |Q(N)|$. Hence the first integral is $< |Q|$ and the two together give $2e^{-4\beta(n+1)p}|Q|$. Note this uses Lemma 2 only in the degenerate case $I \equiv 1$, which is why "say" appears in the original.

6.5 Summing, and the sign that must change. We conclude

$$\begin{aligned}
\int_{V_{n+1}} |\partial_n u_{n+1}|^p &= \int_{\bigcup\{Q \text{ type 1}\}} + \int_{\bigcup\{Q \text{ type 2}\}} \\
&\le e^{-\frac{3}{2}p\beta}\int_{\bigcup\{Q \text{ type 1}\}} |\partial_n u_n|^p + C\,e^{-4p\beta(n+1)} \sum_{Q \text{ type 2}} |Q| \\
&\le A^p e^{-p\beta(n+1)}\cdot e^{-\frac{1}{2}p\beta} + C\,e^{-4p\beta(n+1)} \\
&= A^p e^{-p\beta(n+1)}\Big(e^{-\frac{1}{2}p\beta} + CA^{-p}e^{-3p\beta(n+1)}\Big). \qquad (27)
\end{aligned}$$

The parenthesis term is < 1 if A was large enough.

EXPANSION — a step he compressed, written out

The sign, corrected. [D21] The original prints $e^{+\frac{1}{2}p\beta}$ in the last two lines, and then asserts the parenthesis is < 1. It cannot be: $e^{+\frac{1}{2}p\beta} > 1$ already. The recomputation forces the minus sign. From (25), raising to the p-th power and summing over type 1 cubes,

$$\int_{\bigcup\{Q \text{ type } 1\}} |\partial_n u_{n+1}|^p \leq e^{-\frac{3}{2}p\beta} \int_{V_n} |\partial_n u_n|^p \leq e^{-\frac{3}{2}p\beta} \cdot A^p e^{-p\beta n}$$

by the induction hypothesis (17). Now factor out $A^p e^{-p\beta(n+1)}$:

$$e^{-\frac{3}{2}p\beta} A^p e^{-p\beta n} = A^p e^{-p\beta(n+1)} \cdot e^{-\frac{3}{2}p\beta + p\beta} = A^p e^{-p\beta(n+1)}\, e^{-\frac{1}{2}p\beta}.$$

So the exponent is $-\frac{1}{2}p\beta$, and with it the parenthesis is $e^{-\frac{1}{2}p\beta} + CA^{-p}e^{-3p\beta(n+1)}$. The first summand is < 1 by a fixed margin, and the second is $\leq CA^{-p}$, which is $< 1 - e^{-\frac{1}{2}p\beta}$ once A is large — this is the second constraint on A, the first being the base case of §4.1. Both are lower bounds on A, so one A serves. $\square$

[D20] The step $\sum_{Q \text{ type } 2} |Q| \leq 1$ is just $\bigcup \mathcal{Q}_{n+1} \subset Q(1)$ and $|Q(1)| = 1$.

§7. The conditions on K_n, ε_n, and the conclusion

The K_n and ε_n should satisfy

$$\textbf{(1)} \quad \sum_n \varepsilon_{n+1}^{-d} K_{n+1} e^{-\beta n} < \infty, \tag{28}$$

$$\textbf{(2)} \quad \sum_n \left(K_{n+1}^{-p} + \varepsilon_{n+1}^{\frac{d-1}{2}}\right) \text{ is sufficiently small.} \tag{29}$$

These are clearly compatible: e.g. $\varepsilon_n = C^{-1}n^{-2}$, $K_n = Cn^{2/p}$ for a suitably large constant C.

EXPANSION — a step he compressed, written out

The exponent in (2). [D23] The original prints ε_{n+1}^{d-1}. The quantity condition (2) is used to bound, in §7.3, is $C\sum \varepsilon_{n+1}^{(d-1)/2}$, and since $\varepsilon < 1$ we have $\varepsilon^{d-1} < \varepsilon^{(d-1)/2}$ — so smallness of $\sum \varepsilon^{d-1}$ does *not* give smallness of $\sum \varepsilon^{(d-1)/2}$. The exponent must be $(d-1)/2$, as written above.

Compatibility, checked. [D24] With $\varepsilon_n = C^{-1}n^{-2}$ and $K_n = Cn^{2/p}$:

- (1): the terms are $C^d(n+1)^{2d} \cdot C(n+1)^{2/p} \cdot e^{-\beta n}$, which is (polynomial)$\times$(geometric) and so summable for every C.
- (2): $\sum K_{n+1}^{-p} = C^{-p}\sum(n+1)^{-2} = C^{-p}(\pi^2/6 - 1)$, small for C large; and $\sum \varepsilon_{n+1}^{(d-1)/2} = C^{-\frac{d-1}{2}} \sum(n+1)^{-(d-1)}$, which *converges iff* $d - 1 \geq 2$, and is then small for C large.

So $d \geq 3$ enters here. [A6] For $d = 2$ the second series is $\sum(n+1)^{-1}$, divergent, and this route fails — as it must, since in two dimensions the theorem is false. We record this as *a* place where the hypothesis is used, not *the* place: the imported inequality (7) may also require $d \geq 3$, and since we do not have [1] we cannot say. Reporting the $d = 2$ obstruction as though it were the only one would overstate what this paper shows.

7.1 Convergence. Condition (1) implies, by the Corollary to Lemma 3, that the u_n converge in C^1 norm on the closed upper half space. Call the limit function f.

EXPANSION — a step he compressed, written out

[D25] The original says "the corollary to lemma 2". The Corollary in question is the one stated after Lemma 3 in §5, whose proof invokes Lemmas 1 and 3.

FILLED — an argument omitted in the original, supplied here

From the boundary to the closed half space. [D26, D27] The Corollary (21) bounds $|\nabla u_{n+1} - \nabla u_n|$ *on* $\mathbb{R}^{d-1}$. Two gaps to close.
(i) C^0 *convergence.* From (18), the discs $D(a_Q, \cdot)$ being disjoint,

$$\|u_{n+1} - u_n\|_\infty \le \sup_Q \big|\partial_n u_n(a_Q)\big| \cdot \|F_{\varepsilon_{n+1}}\|_\infty \cdot \frac{\delta_{n+1}}{N} \le 2K_{n+1}e^{-\beta n}\|F_{\varepsilon_{n+1}}\|_\infty \frac{\delta_{n+1}}{N},$$

using (22). Since δ_{n+1} is at our disposal *after* K_{n+1} and ε_{n+1} are fixed, choose it small enough that this is $\le 2^{-n}$; then $\sum \|u_{n+1} - u_n\|_\infty < \infty$ and $u_n \to f$ uniformly on $\mathbb{R}^{d-1}$. (The paper never estimates $u_{n+1} - u_n$ itself, only its gradient.)
(ii) Interior. $v_n := u_{n+1} - u_n$ is harmonic in $\mathbb{R}^d_+$, continuous on $\overline{\mathbb{R}^d_+}$, and tends to 0 at infinity (its boundary data is compactly supported in $Q(1)$). Each $\partial_j v_n$ is harmonic, bounded, and tends to 0 at infinity, so by Background

$$\sup_{\mathbb{R}^d_+} |\nabla v_n| = \sup_{\mathbb{R}^{d-1}} |\nabla v_n| \le CK_{n+1}e^{-\beta n}\varepsilon_{n+1}^{-d}.$$

Condition (28) makes $\sum_n \sup_{\mathbb{R}^d_+} |\nabla v_n| < \infty$, so ∇u_n converges uniformly on $\overline{\mathbb{R}^d_+}$; with (i), $u_n \to f$ in C^1 on $\overline{\mathbb{R}^d_+}$, f is harmonic in $\mathbb{R}^d_+$ (a locally uniform limit of harmonic functions), and $f \in C^1(\overline{\mathbb{R}^d_+})$. This is the C^1-up-to-the-boundary assertion of the Theorem.

We have to show $f = \partial_n f = 0$ on a set of positive measure.

7.2 Where f moved.

$$\big|\{x \in Q(1) : f(x) \neq 0\}\big| \le \sum_n \big|\{x \in Q(1) : u_{n+1}(x) \neq u_n(x)\}\big|, \tag{30}$$

and $\{x : u_{n+1}(x) \neq u_n(x)\}$ is a union of at most $\delta_{n+1}^{-(d-1)}$ discs each of radius $\delta_{n+1}\varepsilon_{n+1}^{1/2}/N$ (because $\operatorname{supp} F_\varepsilon \subset D(0, \varepsilon^{1/2})$). So

$$\big|\{x \in Q(1) : f(x) \neq 0\}\big| \le C \sum_n \delta_{n+1}^{-(d-1)} \Big(\frac{\delta_{n+1}\varepsilon_{n+1}^{1/2}}{N}\Big)^{d-1} \le C \sum_n \varepsilon_{n+1}^{\frac{d-1}{2}}. \tag{31}$$

EXPANSION — a step he compressed, written out

[D28] $u_0 = 0$ on $Q(1)$, so $f \neq 0$ at a point of $Q(1)$ forces $u_{n+1} \neq u_n$ there for some n; that is (30). There are at most $|\mathcal{H}_{n+1}| = \delta_{n+1}^{-(d-1)}$ cubes, hence at most that many discs; each has measure $c_d(\delta_{n+1}\varepsilon_{n+1}^{1/2}/N)^{d-1}$. Multiplying, the δ_{n+1} cancels exactly and $N^{-(d-1)}$ is absorbed into C, leaving $\varepsilon_{n+1}^{(d-1)/2}$ — which is the exponent condition (2) must control, and the reason for the correction at D23.

7.3 Where the normal derivative failed to vanish. Also $\partial_n f = 0$ on $\bigcap_n V_n$. Thus $\big|\{x \in Q(1) : \partial_n f(x) \neq 0\}\big| \le \sum_n |V_n \setminus V_{n+1}|$.

FILLED — an argument omitted in the original, supplied here

Why $\partial_n f = 0$ on $\bigcap_n V_n$. [D30] The paper asserts this without argument. Let $x \in \bigcap_n V_n$. The induction hypothesis (17) gives $\int_{V_m} |\partial_n u_m|^p \leq A^p e^{-p\beta m}$ for every m; since $\bigcap_n V_n \subset V_m$,

$$\int_{\bigcap_n V_n} |\partial_n u_m|^p \leq A^p e^{-p\beta m} \xrightarrow[m\to\infty]{} 0.$$

By §7.1, $\partial_n u_m \to \partial_n f$ uniformly on $\mathbb{R}^{d-1}$, so $\int_{\bigcap_n V_n} |\partial_n u_m|^p \to \int_{\bigcap_n V_n} |\partial_n f|^p$ (the domain has finite measure and the integrands converge uniformly). Hence $\int_{\bigcap_n V_n} |\partial_n f|^p = 0$ and $\partial_n f = 0$ a.e. on $\bigcap_n V_n$ — which is all that is needed, the conclusion being about measure.

If $Q \notin \mathcal{Q}_{n+1}$ but $Q \subset Q' \in \mathcal{Q}_n$, then Q fails (b), i.e.

$$\frac{1}{|Q|}\int_Q |\partial_n u_n|^p > K_{n+1}^p e^{-p\beta n}, \qquad \text{i.e.} \qquad |Q| < K_{n+1}^{-p} e^{+\beta pn} \int_Q |\partial_n u_n|^p.$$

EXPANSION — a step he compressed, written out

[D31] The original prints $e^{-\beta pn}$ here. Rearranging $\frac{1}{|Q|}\int_Q |\partial_n u_n|^p > K_{n+1}^p e^{-p\beta n}$ gives $\int_Q |\partial_n u_n|^p > |Q| K_{n+1}^p e^{-p\beta n}$ and hence $|Q| < K_{n+1}^{-p} e^{+p\beta n}\int_Q |\partial_n u_n|^p$: the exponent flips sign when $e^{-p\beta n}$ moves across. The next display in the original uses $+\beta pn$, confirming the slip.

So

$$|V_n \setminus V_{n+1}| \leq K_{n+1}^{-p} e^{\beta pn} \int_{V_n} |\partial_n u_n|^p \leq A^p K_{n+1}^{-p}, \tag{32}$$

and $\big|\{x \in Q(1) : \partial_n f(x) \neq 0\}\big| \leq A^p \sum_n K_{n+1}^{-p}$.

EXPANSION — a step he compressed, written out

[D32] Summing the previous display over the cubes Q making up $V_n \setminus V_{n+1}$ — they are disjoint and contained in V_n — gives the first inequality; the second is (17), $\int_{V_n} |\partial_n u_n|^p \leq A^p e^{-p\beta n}$, whose exponential exactly cancels the $e^{+\beta pn}$.

7.4 The two costs, added.

$$\big|\{x \in Q(1) : \partial_n f(x) \neq 0\}\big| + \big|\{x \in Q(1) : f(x) \neq 0\}\big| \leq C \sum_n \varepsilon_{n+1}^{\frac{d-1}{2}} + A^p \sum_n K_{n+1}^{-p} < 1, \tag{33}$$

if (2) holds. So $E := Q(1) \cap \{\partial_n f = 0\} \cap \{f = 0\}$ must have positive measure.

EXPANSION — a step he compressed, written out

[D33] $|Q(1)| = 1$, and (33) bounds the measure of the union of the two bad sets by something < 1; the complement within $Q(1)$ is E, so $|E| \geq 1 - (\text{that bound}) > 0$. Note that condition (2) must be small enough to beat the constants C and A^p, both of which are already fixed at this point — which is why (2) says “sufficiently small” rather than “finite”.

7.5 From $\partial_n f = 0$ to $\nabla f = 0$ — the step the paper omits.

FILLED — an argument omitted in the original, supplied here

[D34] The construction has produced $E \subset Q(1)$ with $|E| > 0$, $f|_E = 0$ and $\partial_n f|_E = 0$. The Theorem claims $\nabla f = 0$ on a set of positive measure. The remaining step — absent from the original — is that the *tangential* gradient also vanishes.

Write $\nabla f = (\nabla_T f, \partial_n f)$ on the boundary. We know $\partial_n f = 0$ a.e. on E. For the tangential part: f restricted to $\mathbb{R}^{d-1}$ is C^1 (§7.1) and vanishes identically on E, so by Background , $\nabla_T f(x) = 0$ at every density point x of E. Almost every point of E is a density point, so $\nabla_T f = 0$ a.e. on E.
Hence $\nabla f = 0$ a.e. on E, and discarding a null set we obtain a set of positive measure on which f and ∇f both vanish. Together with §1 ($f \not\equiv 0$, since $f = u_0 \not\equiv 0$ off $Q(1)$) and §7.1 (f harmonic, C^1 up to the boundary), this is the Theorem. ■

§8. Remarks (theirs)

Remark (1). To minimize technicalities we did not try to obtain a Hölder estimate on the gradient although this should be possible along the work of [2].

Remark (2). The main question remains open, that is, whether it is possible to have harmonic functions C^2 or C^∞ smooth up to the boundary whose gradients vanish on sets of positive measure. A variant of the previous construction permits to get a $C^{2+\varepsilon}$ harmonic function $f : \mathbb{R}^d_+ \to \mathbb{R}$ such that f and $D^2 f$ vanish on a common boundary of positive measure.

OUR NOTATION — not in the original

The original prints "C^2 of C^∞" for "or". [E1] The $C^{2+\varepsilon}$ variant of Remark 2 is asserted with neither proof nor reference; we reproduce it as an assertion of the authors and do not verify it. [E2]

§9. Editorial remarks — everything later than the paper

Quarantined here, per the blueprint: nothing in this section was used above.

(1) **The references.** [1] is cited as "private communication (to appear)" and [2] as "to appear". We have not identified the published form of either from a source we have read, and do not guess. [E4, E5] Identifying [1] is the single highest-value follow-up for this digestion: it is the only input we could not verify.

(2) **Preprint versus journal version.** This digestion follows IHES/M/89/31 (September 1989). The published version is Colloq. Math. 60/61 (1990), 253–260, which we could not obtain (the publisher's server was unreachable). Numbering and text may differ. [F4]

(3) **The five slips.** Listed in §0. All are typographical or sign errors; none affects the theorem. Two further points required a real repair rather than a correction: the lower bound for $|1+\partial_n G_\varepsilon|$ near $|x| = 1$ (§2.6, §3.3), and the missing final step §7.5.

(4) **What remains conditional.** The Alexandrov–Kargaev inequality (7). With it, the theorem as stated is proved here in full.

Original: IHES/M/89/31, September 1989; Colloq. Math. 60/61 (1990), 253–260